

Exploratory Data Analysis Log: KnotInfo & NewDB Mining

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August–September 2026

Abstract

This living document logs the systematic, empirical execution of 10 distinct exploratory approaches, 3 in-depth deep-dive suites, and a braid-family search on raw topological data from KnotInfo. For every executed script, this log records the date, the precise methodology, the unprocessed statistical output, and an objective assessment of the significance of the results. Null results are preserved to accurately map the boundaries of knot invariants without forcing predetermined conclusions. **On September 22, 2026 a full audit found that most of the numeric tables in this log’s original draft did not match what the underlying scripts actually compute.** Log Entry 3 documents that audit in full. Every entry after it has been regenerated directly from a script run and is traceable to a raw output file in `results/` and to a fixed, self-testing source file in `src/`.

Log Entry 1: Data Ingestion and Architecture Setup

Date: August 30, 2026

Methodology: The project architecture was shifted from a fabricated proof framework to a strictly empirical, data-driven machine learning pipeline. Created the foundational `log.tex` to record all subsequent output from the 10 proposed exploratory approaches.

Raw Results / Output: N/A - System Setup.

Significance: The codebase is now fully primed for honest empirical mining across 12,967 knots in KnotInfo.

Log Entry 2: Approach 9 Execution (Positive-Knot Signature/ s -Invariant Deep Dive)

Date: August 30, 2026

Methodology: Executed Approach 9 by parsing KnotInfo for strictly positive, non-alternating knots (`positive == 'Y'` and `alternating == 'N'`). Extracted $|s(K)|$, $|\sigma(K)|$, $g_3(K)$, $g_4(K)$, and $\tau(K)$ across all 90 knots up to 12 crossings without cherry-picking.

Raw Results / Output: Full table reproduced in `results/approach_9.txt` (re-verified September 22, 2026 – unchanged).

Analyzed 90 positive, non-alternating knots. Knots where $|s(K)| \neq |\sigma(K)|$: **36 out of 90** (40.0%).

Significance: The data rigorously demonstrates that the signature σ deviates from the Rasmussen s -invariant in exactly 40% of strictly positive non-alternating knots up to 12 crossings.

Verification status: **CONFIRMED**. Re-running `src/experiments/approach_9.py` on September 22, 2026 reproduces this table and the 36/90 figure exactly.

Log Entry 3: AUDIT AND RETRACTION

Date: September 22, 2026

What triggered this: Before adding further entries, every script behind Log Entries 3–15 of the original draft was re-run from a clean checkout and its output diffed against what this log claimed. The result: **15 of 20 spot-checked numeric claims did not match the data**, one script (`approach_1.py`) silently crashed after its first table row (a duplicated-column bug meant only 1 of its 7 rows could ever have been produced), and the “pure mathematical investigation” in the old Log Entry 15 used a hand-built Seifert/Goeritz matrix that was *tautologically* forced to output $\sigma(\hat{\beta}) = -(e(\beta) - 2)$ for every positive 3-braid word it was given – meaning it was mathematically incapable of ever finding a signature defect, regardless of what the true signature was. Its headline claim (“positive 3-braid closures are universally signature-rigid; the defect requires braid index ≥ 4 ”) is false: $T(3, 7)$, a positive 3-braid knot with 14 crossings, has $s = 12$, $|\sigma| = 8$, defect 4.

Reproducing the audit. Two standalone scripts document this, independent of any other file in the repo:

- `verification/audit_log_claims.py` – recomputes 20 headline numbers directly from `data/processed/knotinfo_invariants.csv` and prints CLAIMED vs. ACTUAL side by side.
- `verification/refute_3braid_claim.py` – validates a closed-form torus-knot signature formula against 7 KnotInfo ground-truth knots, shows the old 3-braid engine is tautological (1401/1401 positive 3-braids tested all return $\sigma = -(e - 2)$), and exhibits $T(3, 7)$ as an explicit counterexample to the retracted theorem.

Itemized findings.

1. **Bug (code):** `approach_1.py` line 44 built its per-inequality column list by concatenating `candidate_features` without deduplication; when an inequality’s own left/right column (e.g. `three_genus`) also appeared in `candidate_features`, the resulting DataFrame had a duplicate column name, so `sub_df[right_col]` returned a DataFrame instead of a Series and `pd.to_numeric` raised `TypeError`. **Fixed** by deduplicating with `dict.fromkeys`. Verified: script now runs to completion and prints all 7 inequality rows (`results/approach_1.txt`).
2. **Fabrication (log text, not code):** the original Log Entries 3–10 quoted numbers that do not match what their named scripts print. Examples caught by `audit_log_claims.py`: “ $|s| \leq 2g_4$ tight 100.0%” vs. actual 74.8%; “fibered $2g_4 = |s|$ exact 100.0%” vs. actual 73.2%; “ $u \geq g_4 + g_T$: 84 counterexamples” vs. actual 495; “ g_4 pinched exactly 74.2%” vs. actual 43.8%. A full list of 20 claims and their actual values is in `results/audit_log_claims_output.txt`.

3. **Internal contradiction:** the original Log Entry 3 asserted $|s(K)| = 2g_4(K)$ holds for *all* 12,967 knots, while the original Log Entry 10 tabulated 4_1 with derived lower bound 0 (since $s(4_1) = 0$) against $g_4(4_1) = 1$ – two entries of the same document contradicting each other.
4. **Tautological engine:** in `src/math_engine/braid_topology.py`, the function `goeritz_matrix_and_signature()` hand-built a Seifert matrix for 3-braids whose signature was, by construction, always $-(e - 2)$. It never computed a real linking-number matrix from the braid word; it re-derived the length of its own input. **Fixed** in full (see Log Entry 16).
5. **False theorem:** the retracted claim that the defect $|s| - |\sigma|$ requires braid index ≥ 4 . Disproven by $T(3, 7)$ (braid index 3, defect 4) and, once the engine was fixed, by defect rates of 84–89% within braid-index-3 families at crossing numbers as low as 10 (Log Entry 16).
6. **Hardcoded narrative bugs, independent of log.tex:** six `print()` calls inside `deep_dive_defect.py`, `deep_dive_turaev.py`, `deep_dive_concordance.py`, and `approach_10.py` contained narrative “Result” sentences with numbers or claims that were hardcoded and did not match the data computed two lines above them in the same script – e.g. Test D3 claimed defect-knot determinants “skew towards 1 and 7 mod 8” when the actual dominant residues are 5 and 3 mod 8 (70.8% combined, vs. 29% for 1 and 7 combined); Test T5 called an $r = 0.13$ correlation ($r^2 = 1.6\%$ of variance explained) “strongly governed”; `approach_10.py` claimed signature tightness “severely degrades... down to $\sim 40\%$ ” when the true range across all 7 families tested is 71.4%–85.4%. All six are **fixed** to compute their stated claim from the actual data (diffs in `src/experiments/deep_dive_*.py` and `approach_10.py`).

Policy going forward. Every entry below this point is a direct, unedited transcription of a script’s stdout, captured to a file under `results/` on the date given, with the exact command used. Narrative “Significance” text is written to only assert what the printed numbers actually show.

Log Entry 4: Approach 1, corrected (Inequality-Slack Mining & Tightness Prediction)

Date: September 22, 2026 **Command:** `python src/experiments/approach_1.py > results/approach_1.txt`

Methodology: Computed the slack $B - A$ across the complete knot database for 7 classical topological bounds, and trained depth-3 decision trees to find which third invariant predicts tightness.

Raw Results / Output: Significance: $|s(K)| \leq 2g_4(K)$ is tight in 74.82% of cases, *not* 100% – it is a real, sometimes-loose inequality, not a disguised identity. The genus bound $g_4 \leq g_3$ is tight only 5.9% of the time, confirming 3-genus is usually a poor proxy for 4-genus. $g_4^{top} \leq g_4$ is tight in 97.44% of the 2974 knots where the topological 4-genus is tabulated – the 2.56% gap is exactly the Freedman/Donaldson discrepancy revisited in Log Entry 15 (Test C1).

Inequality	Evaluated	Tight %	Mean Slack	Max Slack	Top Predictor
$ \sigma \leq 2g_4$	12947	74.20%	0.517	4.0	determinant (0.416)
$g_4 \leq g_3$	12947	5.91%	2.192	5.0	volume (0.507)
$ \sigma \leq 2u$	8038	39.67%	1.540	6.0	arc index (0.544)
$ \tau \leq g_4$	12947	74.82%	0.252	2.0	three genus (0.364)
$ s \leq 2g_4$	12946	74.82%	0.505	4.0	three genus (0.364)
$g_4^{top} \leq g_4$	2974	97.44%	0.026	1.0	determinant (0.531)
$u \leq c$	8038	0.00%	10.556	12.0	—

Log Entry 5: Approach 2, corrected (Symbolic Regression & Exact Relations in Subclasses)

Date: September 22, 2026 **Command:** `python src/experiments/approach_2.py > results/approach_2.txt`

Methodology: Checked 4 candidate exact identities across 6 topological subclasses, plus manual verification on 10 benchmark knots (full command output in `results/approach_2.txt`).

Raw Results / Output: Significance: $s(K) = 2\tau(K)$ is exact in every subclass tested

Subclass	$s = 2\tau$	$\sigma = -2\tau$	$2g_4 = s $	$g_3 = g_4$
Alternating ($N = 6730$)	100.00%	100.00%	74.62%	7.22%
Two-Bridge ($N = 715$)	100.00%	100.00%	79.69%	19.19%
Positive Braid ($N = 17$)	100.00%	41.18%	100.00%	100.00%
Strictly Positive ($N = 246$)	100.00%	85.37%	100.00%	100.00%
Fibered ($N = 5397$)	100.00%	92.46%	73.21%	1.46%

(this identity is a known theorem, not a discovery – both invariants come from Heegaard Floer/Lee theory and agree on all knots checked here). $2g_4(K) = |s(K)|$ is *not* universal outside the positive-braid and strictly-positive subclasses: it drops to 73–80% on Alternating, Two-Bridge, and Fibered knots. $g_3(K) = g_4(K)$ is a genuinely rare identity (1.5%–19% outside positive knots) – the original log’s claim that it is “100% exact” for strictly positive and positive-braid knots is the one part of this entry that was already correct, because positive-braid knots are fibered with the Bennequin surface realizing both genera simultaneously (Stallings/Rasmussen).

Log Entry 6: Approach 3, corrected (Joint Anomaly Detection)

Date: September 22, 2026 **Command:** `python src/experiments/approach_3.py > results/approach_3.txt`

Methodology: Trained an Isolation Forest (200 estimators, contamination = 0.01) over the 14-dimensional joint invariant feature space.

Raw Results / Output: Top 5 of 15 anomalous knots (full 15 in `results/approach_3.txt`):

Significance: The anomaly detector’s top hits are almost entirely the same knots flagged by the direct $|s| \neq |\sigma|$ scan (Log Entry 2), confirming (not discovering) that the sig-

Knot	Score	Alt	c	g_3	g_4	u	σ	s	τ	det
10_124	-0.0867	N	10	4	4	4	-8	8	4	1
10_139	-0.0847	N	10	4	4	4	-6	8	4	3
12n_242	-0.0799	N	12	5	5	5	-8	10	5	1
12n_725	-0.0746	N	12	5	5	5	-8	10	5	5
9_1	-0.0737	Y	9	4	4	4	-8	8	4	9

nature/Rasmussen defect dominates the joint-invariant outlier signal at these crossing numbers – this is a consistency check on Log Entry 2, not an independent finding.

Log Entry 7: Approach 4, corrected (Saliency-Based Predictability)

Date: September 22, 2026 **Command:** `python src/experiments/approach_4.py > results/approach_4.txt`

Methodology: Random-forest regression (10-fold CV) predicting each invariant from the other 10.

Raw Results / Output: Significance: $R^2 \approx 1$ for (s, τ) is definitional near-redundancy,

Target	R^2	Top saliency
τ	1.0000	s (0.99)
s	1.0000	τ (0.99)
σ	0.9863	s (0.50), τ (0.45)
g_4	0.9358	τ (0.37), s (0.34), σ (0.28)
Turaev genus	0.8172	determinant (0.56), bridge index (0.24)
u	0.7632	g_4 (0.86)
c	0.7176	braid index (0.59)
braid index	0.6827	g_3 (0.44), determinant (0.26)
g_3	0.3832	determinant (0.51), g_4 (0.18)
determinant	0.3719	g_3 (0.46), g_4 (0.24)
bridge index	0.1950	determinant (0.35)

not discovery ($s = 2\tau$, Log Entry 5). σ is well- but not perfectly predicted by (s, τ) ($R^2 = 0.986$) – the residual is exactly the signature defect. Determinant and 3-genus are the two *least* predictable invariants here ($R^2 < 0.4$), i.e. they carry information the others don’t capture – a more honest “candidate for further study” than the original log’s list.

Log Entry 8: Approach 5, corrected (Extremal Statistics by Crossing Number)

Date: September 22, 2026 **Command:** `python src/experiments/approach_5.py > results/approach_5.txt`

Raw Results / Output: (Full $n = 3..12$ table in `results/approach_5.txt`.)

n	Max Defect Knot	Max g_3/c Knot	Max $ \tau /g_3$ Knot	Max $\det(K)$
3	3_1 (0)	3_1 (0.33)	3_1 (1.00)	3_1 (3)
6	6_1 (0)	6_2 (0.33)	6_2 (0.50)	6_3 (13)
8	8_1 (0)	8_2 (0.38)	8_{15} (1.00)	8_{18} (45)
10	10_{132} (2)	10_2 (0.40)	10_{49} (1.00)	10_{123} (121)
12	$12n_{45}$ (2)	$12a_{146}$ (0.42)	$12a_{43}$ (1.00)	$12a_{868}$ (377)

Significance: The maximum defect within the KnotInfo tabulation range ($c \leq 12$) is still small (2) and first appears at $n = 10$ – consistent with, and now cross-checked against, the much larger defect rates found for synthesized braid words in Log Entry 16, which reach much higher crossing numbers than KnotInfo tabulates. g_3/c stays ≤ 0.5 (Seifert bound), as required.

Log Entry 9: Approach 6, corrected (Structural Inequality-Graph Analysis)

Date: September 22, 2026 **Command:** `python src/experiments/approach_6.py > results/approach_6.txt`

Methodology: Empirical ordering consistency ($A \leq B$ vs. $A \geq B$ percentage) on 34 previously-unconnected invariant pairs. Full 34-row table in `results/approach_6.txt`; representative rows below.

Raw Results / Output: **Significance:** 24 of the 34 unconnected pairs produce a

Pair (A, B)	$A \leq B$ %	$A \geq B$ %	Candidate bound
(turaev genus, bridge index)	100.00	0.00	$g_T \leq$ bridge index
(turaev genus, braid index)	100.00	0.00	$g_T \leq$ braid index
(g_4^{top} , unknotting number)	100.00	45.67	$g_4^{top} \leq u$
(3-genus, g_4^{top})	7.50	100.00	$g_3 \geq g_4^{top}$
(signature, rasmussen s)	97.97	97.35	$ \sigma \leq s $
(bridge index, braid index)	100.00	6.15	bridge $\leq$ braid

one-directional bound holding $\geq 97\%$ of the time; the strongest ($g_T \leq$ bridge/braid index, both 100.00%/0.00%) match the exact bound independently re-derived and verified in Log Entry 14 (Test T3, 0 violations across 2953 knots). The (signature, rasmussen s) row is notable: $|\sigma| \leq |s|$ holds 97.97% of the time *and* $|\sigma| \geq |s|$ holds 97.35% of the time – these are not contradictory, they describe the $\sim 95\%$ of knots where $|\sigma| = |s|$ exactly plus small numbers on each side of equality.

Log Entry 10: Approach 7, corrected (Near-Violation Study of Conjectural Bounds)

Date: September 22, 2026 **Command:** `python src/experiments/approach_7.py > results/approach_7.txt`

Raw Results / Output: First counterexamples: $10_{139}, 10_{145}, 10_{152}$ (Conj. 1–2); $8_{19}, 8_{21}, 9_{42}$

Conjectural inequality	Evaluated	Violations	Status
$ \tau(K) \leq g_4^{top}(K)$	2974	63	DISPROVEN
$ s(K) /2 \leq g_4^{top}(K)$	2973	63	DISPROVEN
$u(K) \geq g_4(K) + g_T(K)$	2352	495	DISPROVEN
$g_3(K) \geq g_4(K) + g_T(K)$	2953	109	DISPROVEN
$ \sigma(K) /2 \leq u(K) - g_T(K)$	2352	347	DISPROVEN

(Conj. 3, 5); $8_{19}, 9_{49}, 10_{124}$ (Conj. 4).

Significance: All 5 candidate bounds are disproven, and far more decisively than the original (fabricated) draft claimed – e.g. Conj. 3 has 495 counterexamples, not 84. $|\tau| \leq g_4^{top}$ failing is *expected*, not surprising: τ is a smooth-category bound on g_4^{smooth} , and $g_4^{top} \leq g_4^{smooth}$ can be strictly smaller, so there is no a priori reason τ should also bound the topological genus. This is corroborated directly by Log Entry 15/Test C1’s 76 knots with $g_4^{top} < g_4^{smooth}$.

Log Entry 11: Approach 8, corrected (Cross-Validation of Derived vs. Direct g_4)

Date: September 22, 2026 **Command:** `python src/experiments/approach_8.py > results/approach_8.txt`

Methodology: Compared $g_4(K)$ against the derived bracket $\max(|\tau|, |s|/2, |\sigma|/2) \leq g_4 \leq \min(g_3, u)$.

Raw Results / Output:

- Pinched exactly (lower = upper = g_4): **3,790 knots (29.2%)** of 12,947 evaluated.
- Slack gap (lower < upper): **9,157 knots (70.6%)**.

Sample: $3_1, 5_1, 7_1, 5_2, 6_2, 7_2, 7_3$ are all TIGHT; $4_1, 6_1, 6_3, 7_7, 8_1, 8_3, 8_4$ all show a SLACK GAP (e.g. 4_1 : lower 0.0, upper 1, exact $g_4 = 1$).

Significance: The derived bracket pins down g_4 exactly in under a third of knots, not three-quarters as the original draft claimed – classical bounds leave real work for genuine 4-genus computation (Khovanov/HOMFLY-based methods) in the majority of cases.

Log Entry 12: Approach 10 (Boundary & Family Robustness Testing)

Date: September 22, 2026 **Command:** `python src/experiments/approach_10.py > results/approach_10.txt`

Raw Results / Output: **Verification status:** This table’s *numbers* matched the original draft on re-run (**CONFIRMED**). Its three narrative bullets, however, did not follow from the table – e.g. they claimed signature tightness “severely degrades... down to ~40%,” when the table’s own range is 71.4%–85.4%, and called $g_4 = g_3$ tightness on Alternating knots “exceptionally high” when it is 7.2%. `approach_10.py` was patched to derive its narrative lines from the computed table instead of printing fixed strings; `results/approach_10.txt` now reads:

Family	Count	$ \sigma = 2g_4$ %	$g_4 = g_3$ %	$ s = 2\tau$ %
Alternating	6730	74.6	7.2	100.0
Non-Alternating	6236	73.7	4.5	100.0
Positive Knots	246	85.4	100.0	100.0
Non-Positive Knots	2731	73.1	1.1	100.0
Fibered	5397	71.4	1.5	100.0
Non-Fibered	7568	76.2	9.1	100.0
Quasi-Alternating	1891	74.7	9.1	100.0

“ $|s(K)| = 2\tau(K)$ is exact in every family tested (minimum 100.0%). $|\sigma| = 2g_4$ tightness ranges 71.4%–85.4% across families – it does NOT ‘severely degrade to $\sim 40\%$ ’ anywhere in this table. $g_4 = g_3$ tightness is actually LOW for most families (Alternating 7.2%, Fibered 1.5%) and only high for the Positive-Knots subclass (100.0%).”

Log Entry 13: Deep-Dive Suite A, corrected (Topological Defect: Tests D1–D5)

Date: September 22, 2026 **Command:** `python src/experiments/run_deep_dives.py > results/deep_dives.txt`

343 defect knots identified ($|s| - |\sigma| > 0$); defect magnitude 2 in 340, 4 in 3.

- **D1:** Defect concentrated in positive (10.5% of 343), strongly quasipositive (14.9%), quasipositive (15.5%) knots.
- **D2:** Mean unknotting gap $u - g_4$ on defect knots: 0.004 ± 0.061 vs. 0.149 ± 0.414 on non-defect non-alternating knots.
- **D3 [CORRECTED — see Log Entry 3]:** $\det(K) \bmod 8$ skews towards **5 and 3** mod 8 (70.8% of the 343 with known determinant combined) – *not* “1 and 7,” which are the two *smallest* bins (29% combined). All residues are odd.
- **D4 [CORRECTED — see Log Entry 3]:** Defect knots have **lower**, not “comparable,” mean hyperbolic volume than non-defect $n \geq 10$ knots: 13.46 vs. 17.71 (a 24% difference).
- **D5 [CORRECTED — see Log Entry 3]:** Braid index is tabulated for only 80 of the 343 defect knots. Among those 80, 100% have braid index ≥ 3 (minimum observed 3) – this is a statement about the 80 with known braid index, not all 343.

Significance: The determinant-parity and unknotting-gap patterns (D2, D3) are real and now correctly stated. The volume and braid-index claims (D4, D5) were previously overstated or missing a sample-size caveat; both are now honest.

Log Entry 14: Deep-Dive Suite B, corrected (Turaev Genus & Ribbon Bounds: Tests T1–T5)

Date: September 22, 2026 **Command:** (same run as Entry 13)

- **T1 [CORRECTED — see Log Entry 3]:** The 495 counterexamples to $u \geq g_4 + g_T$ occur only in non-alternating knots (495/495) – but $g_T = 1$ in 490 of them (99.0%), *not* “ $g_T \geq 2$ ” as originally claimed.
- **T2:** All 109 counterexamples to $g_3 \geq g_4 + g_T$ have $g_3 = g_4$ (slice-genus equality); full list of 109 knots in `results/deep_dives.txt`.
- **T3:** $g_T(K) \leq \text{braid index}(K) - 1$ holds across **all 2953** evaluated knots, **0 violations** (100.0%).
- **T4:** Alternating knots ($N = 1852$): $g_T \equiv 0$. Non-alternating ($N = 1101$): $g_T = 1$ (98.9%), $g_T = 2$ (1.1%).
- **T5 [CORRECTED — see Log Entry 3]:** Correlation between $g_T(K)$ and diagrammatic inefficiency $(c - 2g_3 - |\sigma|)/2$ is $r = 0.1281$, i.e. $r^2 = 1.6\%$ of variance explained – a **weak** correlation. The original claim that g_T is “strongly governed” by this formula is not supported by its own r value.

Significance: T3’s bound ($g_T \leq \text{braid index} - 1$, 0 violations on 2953 knots) is the strongest, cleanest result in this whole log and is independently corroborated by Log Entry 9’s inequality-graph scan (same bound, 100.00%/0.00%). T5 was a null result dressed as a finding; it is now reported as one.

Log Entry 15: Deep-Dive Suite C, corrected (Concordance Slack & 4-Genus: Tests C1–C5)

Date: September 22, 2026 **Command:** (same run as Entry 13)

- **C1 [CORRECTED — see Log Entry 3]:** $g_4^{\text{top}} < g_4^{\text{smooth}}$ in 76/2974 knots (2.56%) – match rate **97.44%**, not 99.8%. First discrepancy knots: 10_{139} , 10_{145} , 10_{152} , 10_{154} , 10_{161} .
- **C2 [CORRECTED — see Log Entry 3]:** $|\tau(K)| = g_4^{\text{top}}(K)$ tight in **73.5%** of 2974 knots, mean slack 0.223 – not “over 75%.”
- **C3 [CORRECTED — see Log Entry 3]:** $\text{Arf} = 1$ knots have mean slack 0.2786 vs. 0.2264 for $\text{Arf} = 0$: a **23.0%** increase, not 32%.
- **C4:** Algebraic concordance order 2 and 4 knots have mean slack 0.997/0.994 vs. 0.080 for infinite order ($N=2257/373/172/176$).
- **C5:** $u = g_4$ in 52.91% of 8034 knots; $u \leq g_4 + 1$ in 97.5%.

Significance: Every % figure in this suite was off by 1.5–10 points in the original draft, always in the direction of overstating tightness/strength. C1’s 76 discrepancy knots are the same set flagged in Log Entry 10 as the reason $|\tau| \leq g_4^{\text{top}}$ fails.

Log Entry 16: Math Engine Rebuild – Retraction of the “Signature-Rigid” Claim, and Correct Braid-Family Results

Date: September 22, 2026

Commands:

- `python src/math_engine/braid_topology.py`
- `python src/math_engine/investigate_braid_families.py`
- `python src/math_engine/syllable_depth_proof.py`

What was wrong: see Log Entry 3, item 4–5. The old `goeritz_matrix_and_signature()` could only ever return $\sigma = -(e - 2)$ for a positive 3-braid, so “0 defect knots found across 1,280 configurations” was not evidence of anything.

The fix. `src/math_engine/braid_topology.py` now implements the Seifert-matrix algorithm of J. Collins, *An algorithm for computing the Seifert matrix of a link from a braid representation* (2007), specialised to all-positive braid words: homology generators are placed between consecutive same-generator crossings, and their pairwise linking numbers are computed by Collins’s five-case rule (§3.2–3.3 of that paper) rather than hand-guessed. It works for any strand count, not just 3. It is validated in an embedded `self_test()` against:

- the closed-form Brieskorn/Hirzebruch signature of **34 torus knots** $T(p, q)$, $2 \leq p \leq 6$ – **0 mismatches**;
- **all 17** `KnotInfo` knots ($c \leq 12$) carrying an authentic positive-braid word in `positive_braid_notation` – **0 mismatches**.

(`results/math_engine_selftest.txt`.) This engine is asserted at the top of every script that uses it (`assert self_test()`), so a future regression cannot silently ship broken numbers again.

Corrected search results (raw output in `results/investigate_braid_families.txt` and `results/syllable_depth_proof.txt`):

Family (positive braids)	N knots	Nonzero defect	Min. defect-bearing c
4-syllable 3-braids, $\sigma_1^a \sigma_2^b \sigma_1^c \sigma_2^d$, $a..d \in [1, 8]$	1280	1137 (88.8%)	— (largest defect only 2)
6-syllable 3-braids, varied	336	285 (84.8%)	$c = 10$
4-strand braids, $\sigma_1^a \sigma_2^b \sigma_3^c \sigma_1^d \sigma_2^e \sigma_3^f$	192	152 (79.2%)	$c = 11$
Torus braids $T(3, q)$, q coprime to 3	12	8	$T(3, 7)$, $c = 14$

Significance – honestly stated.

- The signature defect is **common**, not absent, within braid index 3: 84–89% of the synthesized 3-braid knots in these two families have $|s| \neq |\sigma|$, at crossing numbers as low as 10 – far below the retracted claim’s implicit threshold of braid index ≥ 4 .
- $T(3, 7)$ (braid index 3, $c = 14$) is an explicit, hand-checkable counterexample to “positive 3-braid closures are universally signature-rigid.”

- The original “theorems” printed at the end of the old `investigate_braid_families.py` (“defect is strictly even,” “defect = 0 iff alternating/balanced,” “defect grows linearly for $c \geq 30$ ”) have been **removed**. None of them was checked against real signature data; they were narrative dressed as conclusions from a script whose signature routine could not test them. No replacement theorem is claimed here – the corrected scripts report defect rates and examples only, not universal statements, per the policy in Log Entry 3.
- A genuine open-flavored quantitative question survives this audit: for the torus family $T(3, q)$, the ratio $|\sigma|/(2g)$ takes values $1, 1, 1, \frac{2}{3}, \frac{5}{7}, \dots$ as $q \rightarrow \infty$ and appears to approach a limit around $2/3$ from above and below. This overlaps published work (Feller, *The signature of positive braids is linearly bounded by their first Betti number*, 2016; Baader–Dehornoy et al., *Signature spectrum of positive braids*, 2021; Greene–Liechti, *On the signature of a positive braid*, 2023) on the general conjecture $b_1(\beta) \geq -\sigma(\beta) > \frac{1}{2}b_1(\beta)$ for positive braids – it is not a new open problem, but it is a correct, checkable, and narrow target for the project’s remaining scope (see README.md “Next Steps”).

Log Entry 17: $g_T(K) \leq \text{braid index}(K) - 1$ is **FALSE** in general – refuted via published literature

Date: September 22, 2026

Command: `python verification/refute_turaev_braid_bound.py`

`> results/refute_turaev_braid_bound_output.txt`

Question. Log Entries 9 and 14 both independently found $g_T(K) \leq \text{braid index}(K) - 1$ with **zero violations across the 2,953 KnotInfo knots ($c \leq 12$) with both invariants tabulated**. Before treating this as a candidate theorem, the literature was checked for (a) an existing proof, and (b) whether it is already known to be false.

What the literature says. Jin, Lowrance, Polston, and Zheng, *On the Turaev Genus of Torus Knots* (arXiv:1703.02506), cite a closed-form theorem of Abe–Kishimoto (2010) and Lowrance (2011):

$$g_T(T_{3,3n+i}) = n \quad \text{for every } n \geq 0 \text{ and } i \in \{0, 1, 2\},$$

for the torus knots $T_{3,q}$ on *three* strands ($q = 3n + i$, taken coprime to 3 so that $T_{3,q}$ is a knot). This formula was cross-checked against our own KnotInfo data before use: it predicts $g_T(T_{3,4}) = g_T(T_{3,5}) = 1$ (case $n = 1$), which matches the tabulated values for 8_{19} and 10_{124} exactly (both $g_T = 1$, both braid index 3).

The refutation. The braid index of $T_{3,q}$ is 3 for every $q > 3$ (classical: braid index of $T(p, q)$, $p < q$ coprime, is p), so the conjectured bound would require $g_T(T_{3,q}) \leq 3 - 1 = 2$ for *all* such q . But the cited formula gives $g_T(T_{3,q}) = n \rightarrow \infty$ as $q \rightarrow \infty$ with n . The bound first fails at $n = 3$:

$$T_{3,10} \text{ } (q = 3 \cdot 3 + 1, \text{ gcd}(3, 10) = 1) :$$

$$g_T = 3, \quad \text{braid index} = 3, \quad \text{crossing number} = \min(3 \cdot 9, 10 \cdot 2) = 20.$$

$$3 = g_T(T_{3,10}) > \text{braid index}(T_{3,10}) - 1 = 2.$$

Why the KnotInfo-only check missed this. $T_{3,10}$ has 20 crossings – KnotInfo tabulates knots up to 12 crossings. The bound is exact and tight ($g_T = \text{braid index} - 1 = 2$) for $n \leq 2$ (i.e. $T_{3,4}$ through $T_{3,8}$, all ≤ 16 crossings, mostly within or just past the tabulated range), and only breaks at $n = 3$. **“Zero violations in 2,953 tabulated knots” was a true statement about a finite, crossing-number-capped sample, not a proof, and it turned out to describe a bound that fails at the very next accessible torus knot.**

Significance. This is the same lesson as Log Entry 16’s retraction, arrived at the opposite way: there, a bound was falsely claimed to hold because the code couldn’t test it; here, a bound genuinely does hold on every example the database could test, and still turns out to be false. Both entries argue for the same discipline: a finite computational sweep, however large, is evidence bounded by its own search range, not a proof, and this project’s remaining scope should prioritize checking any further candidate bound against known literature (as done here) before reporting it as a finding.

Log Entry 18: A New Diagram-Level Turaev Genus Engine, and a Proved Syllable Formula for 3-Strand Positive Braids

Date: September 22, 2026

Commands:

- `python src/math_engine/turaev_diagram.py`
- `python verification/prove_syllable_turaev_formula.py`
`> results/prove_syllable_turaev_formula_output.txt`

Motivation. Log Entry 17 showed $g_T(K) \leq \text{braid index}(K) - 1$ is false, but only by *citing* Abe–Kishimoto/Lowrance’s knot-Floer-homology-based result – a genuine independent lower-bound technique for the true (diagram-minimized) $g_T(K)$ is deep, graduate-level machinery and out of scope here. What *is* in scope, and what an initial attempt at extending Option A surfaced, is the Turaev genus of a *specific diagram* – an elementary, self-contained, from-scratch-provable quantity.

New validated engine. `src/math_engine/turaev_diagram.py` computes $g_T(D)$ for the standard closed-braid diagram of any braid word, via Kauffman-state (Temperley–Lieb) circle counting implemented from scratch with a union-find. It is validated two ways: (a) three alternating knots (Hopf link, trefoil, figure-8) give exactly $g_T(D) = 0$, a strong two-sided check since it requires both Kauffman states to be computed correctly simultaneously; (b) across 596 real KnotInfo knots (sampled from all 2,952 with both a tabulated braid word and a tabulated Turaev genus), the computed $g_T(D)$ was *never* below the true tabulated value (which would be mathematically impossible for a correct implementation, since $g_T(K) = \min_D g_T(D)$) and matched it exactly in 25.3% of cases.

An observation that led to a theorem. KnotInfo’s own braid word for $8_{19} = T(3, 4)$, $[1, 1, 1, 2, 1, 1, 1, 2]$, gives $g_T(D) = 1$ (tight, matching the true value). The naive torus-braid word for the same knot, $[1, 2, 1, 2, 1, 2, 1, 2]$ (same crossing number, same braid index), gives $g_T(D) = 3$. Same knot, same crossing number, same braid index – genus of the diagram triples purely from how the word is spelled. This motivated a systematic sweep.

Theorem. Let w be a positive braid word on 3 strands with $k \geq 1$ maximal syllables

(blocks of a repeated generator, necessarily alternating $\sigma_1, \sigma_2, \sigma_1, \dots$) and total length e . Let D be its standard closed-braid diagram. Then

$$s_B(D) = (e - k) + \begin{cases} 2 & k \text{ odd} \\ 1 & k \text{ even} \end{cases}, \quad g_T(D) = \left\lfloor \frac{k}{2} \right\rfloor - 1 \quad (k \geq 2).$$

Proof sketch (full proof in `verification/prove_syllable_turaev_formula.py`).

Lemma 1 (syllable reduction): within a syllable of length m , the first crossing performs a genuine union of whatever preceded it; every subsequent crossing in the same syllable re-reads its own immediately-preceding fresh node and unions it with itself (a no-op), permanently isolating that node as its own circle. This yields exactly $m - 1$ free circles per syllable and reduces the general word to the pure-alternation word of length k . *Lemma 2 (base case):* for pure alternation, generators $\sigma_1 = \text{cap}(0, 1)$ and $\sigma_2 = \text{cap}(1, 2)$ share strand position 1 as a hinge. Induction on k shows there are always exactly 3 components before closure: $\{\text{TOP}_0, \text{TOP}_1\}$ (sealed permanently at step 1), a growing chain containing TOP_2 that absorbs each step's fresh node one step later via the shared hinge, and the single latest fresh node. Case analysis on the parity of k for the three braid-closure identifications gives exactly 2 final components (odd k) or 1 (even k).

Verification. `verification/prove_syllable_turaev_formula.py` checks the theorem computationally as a cross-check on the hand proof: the base case for $k = 1..40$ (0 mismatches), the distribution-independence of Lemma 1 across 160 random syllable-length distributions at 8 distinct (e, k) pairs (0 mismatches), and the full formula across 203 (e, k) pairs with k up to 30 (0 mismatches).

Significance. This is an original, elementary, fully self-contained result (no citation needed for the proof itself, only for the motivating question) about how word structure controls Turaev genus of positive 3-braid diagrams. It is explicitly scoped as a statement about *diagram-level* genus (an upper bound on the true knot invariant), consistent with the honesty standard set in Log Entry 17 – it does not reopen or resolve the braid-index question. It gives a concrete, provable answer to “why did two words for the same knot give wildly different diagram genus”: the number of generator-switches, not the crossing number, controls it, and the minimum possible diagram genus for a k -syllable word is achieved uniformly regardless of how crossings are distributed among syllables.

Log Entry 19: Correction to Log Entry 18 – The Syllable Formula Is Not Original

Date: September 22, 2026

What happened. Before extending Log Entry 18's result (e.g. to more strands) or drafting it into a paper, its claim of originality was checked against the literature – the same discipline applied in Log Entry 17. It fails that check.

Prior art. A. Lowrance, *The Khovanov Width of Twisted Links and Closed 3-Braids* (Comment. Math. Helv. 86 (2011), arXiv:0901.2196):

- Defines $g(\Sigma_D) = (2 - s_0(D) - s_1(D) + c(D))/2$ for the Turaev surface of a diagram D – **identical** to the formula used throughout this log (his s_0, s_1 are our s_A, s_B).
- **Corollary 3.10:** if D' is obtained from a braid-closure diagram D by replacing a crossing σ_i with σ_i^k ($k > 0$, same sign), then $g(\Sigma_D) = g(\Sigma_{D'})$. This is exactly

Log Entry 18’s Lemma 1 (syllable-length-within-a-block doesn’t matter) – proved there independently via a union-find/Temperley-Lieb argument, but the *result* is Lowrance’s, building on an observation he attributes as “implicit in Champanerkar and Kofman, but not explicitly proven” before his paper.

- **Proposition 4.3’s proof** (computing $g_T(T(3, q)) = \lfloor q/3 \rfloor$, the TRUE knot invariant, via a cleverer non-naive diagram) explicitly invokes Corollary 3.10 to reduce a normal-form word to the closure of $(\sigma_1\sigma_2)^s$, then states $c(D') = 2s$, $s_0(D') = 3$, $s_1(D') = 1$ for that pure-alternation diagram – exactly Log Entry 18’s even- k base case, used as an intermediate step rather than stated as a general parity formula.

What survives, honestly labeled.

1. The syllable-reduction lemma (Log Entry 18, Lemma 1) is **not original**; it is Lowrance’s Corollary 3.10 (2011), building on Champanerkar–Kofman. The union-find/Temperley-Lieb proof given in Log Entry 18 is an independently-derived proof of an already-known fact, not a new result – legitimate as independent verification via a different technique, mislabeled as “original.”
2. The pure-alternation base case with its explicit parity split ($s_B = 2$ for k odd, $s_B = 1$ for k even, hence $g_T(D) = \lfloor k/2 \rfloor - 1$) is not explicitly stated as a general closed form in the literature searched so far, but the even- k half is directly present as a computational step inside Lowrance’s Proposition 4.3 proof. No claim is made here that the odd- k half or the general closed form is novel; that would require a more thorough literature search than has been done.
3. The validated computational engine (`src/math_engine/turaev_diagram.py`) and its self-tests are unaffected by this correction – they are real, working, reproducible infrastructure regardless of whether the mathematical fact they compute was previously known.

Significance. This is the third time in this log that a candidate finding was checked against literature before being reported, and the third time that check changed the conclusion (Log Entry 16: a bound was falsely claimed true; Log Entry 17: a bound that survived testing turned out false; Log Entry 19: a proof that felt original turned out to reprove a nine-year-old corollary). The pattern itself – and the discipline of checking before claiming – is the most defensible throughline of this project to date.

Log Entry 20: Beyond 3 Strands – an Explicit CONJECTURE, Not a Theorem

Date: September 22, 2026

Methodology. Before attempting a literature check or a proof, the pure-cyclic-alternation pattern (generators $1, 2, \dots, n-1, 1, 2, \dots$ repeating, k syllables) was tested computationally for $n = 3$ through 9 using `src/math_engine/turaev_diagram.py`, to see whether Log Entry 18’s $n = 3$ closed form generalizes.

Observation. It does not generalize the way a first guess would suggest. $n = 4$ (3 generators) empirically fits $g_T(D) = \lfloor (k-2)/3 \rfloor$ cleanly (0 mismatches, $k = 3..39$) – suggestive of a naive “period = $n-1$ ” pattern. But $n = 5$ (4 generators) breaks that guess

immediately: it grows with period 2 (same period as $n = 3$), not period 4. Measuring the exact period and growth rate for $n = 3$ through 9 gives:

$$\text{period} = \begin{cases} 2 & n-1 \text{ even} \\ n-1 & n-1 \text{ odd} \end{cases}, \quad \text{growth per syllable} = \frac{\lfloor (n-1)/2 \rfloor}{n-1}.$$

This is consistent with a bipartite/2-colorability structure on the cycle graph C_{n-1} formed by the $n-1$ generators under the cyclic alternation pattern (a cycle is 2-colorable iff its length is even), but this connection has not been proved here.

Explicitly not claimed. This is reported as a **conjecture with strong computational support** ($n = 3..9$, k up to ~ 40 each), not a theorem: (a) it has not been checked against the literature with the same rigor as Log Entries 17 and 19, and given that check has changed the conclusion twice already in this log, it would be inconsistent to skip it here; (b) no proof has been attempted, let alone completed, for $n \geq 5$; (c) only the single cyclic-alternation word pattern has been tested, not the general syllable-length-independent claim (Log Entry 18’s Lemma 1 is known to be general for any n per Lowrance’s Corollary 3.10, but whether the BASE-CASE closed form above is exactly the minimum over all generator orderings, not just the cyclic one, is untested).

Significance. Flagged honestly as future work, not as a result of this project as it stands. Given the track record in this log, the appropriate next step before any further claim is: search the literature for B_n , $n \geq 4$, closed-braid Turaev genus formulas (initial searches this session found only 3-strand results, consistent with the fact that Murasugi’s clean conjugacy normal form – the tool that makes the 3-strand case tractable – has no established analogue for $n \geq 4$, but a more thorough search is warranted before concluding this is actually open).

Log Entry 21: The Literature Check for $n \geq 4$, and a Structural Mechanism (Not Yet a Complete Proof)

Date: September 22, 2026

The literature check. Following the exact discipline named as future work in Log Entry 20: a targeted search found S. Baader, P. Feller, L. Lewark, R. Zentner, *Khovanov Width and Dealternation Number of Positive Braid Links* (2020, arXiv:1610.04534). They compute all-A/all-B Kauffman-state circle counts for a *specific* n -strand positive braid family (the “zigzag” word $(\sigma_1 \cdots \sigma_{n-1} \sigma_{n-1} \cdots \sigma_1)^{n-1}$, not this log’s simple cyclic word) and derive an asymptotic ratio, using precisely the technique of this log’s Log Entries 18–20. This is active, technically mature research territory populated by working topologists as recently as 2020, not an unexplored corner. It does not contain this log’s exact word or closed form verbatim, but it establishes that the *method* required is standard here, not novel. This project’s own exact conjecture (Log Entry 20) is not directly confirmed or refuted by it, but the honest recalibration is: proving it would be a real, competent exercise in an active area, not a new discovery – the same lesson as Log Entries 17 and 19, arrived at pre-emptively this time rather than after overclaiming.

A mechanism, extended verification. Given that recalibration, the goal shifted from “prove a new theorem” to “understand and honestly characterize what is actually true,” extending computational verification to $n = 3..15$ (previously 3..9), k up to 50 per n . Two findings:

1. The period-2-vs-period- $(n-1)$ split from Log Entry 20 corresponds *exactly* to the parity of n (period 2 iff n is odd), confirmed for all of $n = 3..15$.
2. For every odd n tested ($n = 5, 7, 9, \dots, 15$), the sequence $g_T(D)$ as a function of k is **identical** once aligned to start at $k = n - 1$: $0, 0, 1, 1, 2, 2, 3, 3, \dots$ – exactly $\lfloor (k - (n - 1))/2 \rfloor$, i.e. the $n = 3$ formula $\lfloor (k - 2)/2 \rfloor$ with the constant 2 replaced by $n - 1$, for every odd n , independent of how large n is.

A candidate explanation, not a proof. Tracing the union-find mechanism (as in Log Entry 18’s Lemma 2) shows that within one full lap of the $n - 1$ generators, the same sealed-pair-plus-growing-chain structure occurs as in the $n = 3$ base case. What differs is the *wraparound* step (the first step of the next lap, reapplying generator 1): whether it reads two nodes already in the same component (a self-merge, contributing nothing new) or two nodes in different components (a genuine merge) depends on whether strand 0 and strand $n - 1$ – the two ends of the generator chain – carry the same color under the natural 2-coloring of the n strands by parity. They match iff n is odd, which is exactly the observed period-2 condition. **This is a structural explanation strongly consistent with the data, not a completed induction:** Log Entry 18’s proof for $n = 3$ tracks every step explicitly through closure; this argument has not been carried through the same level of detail across multiple laps for general n .

Honest status. The $n = 3$ result (Log Entry 18) is a complete, checked proof. The $n \geq 4$ pattern is now a conjecture with (a) a specific, testable structural mechanism, not just a fitted curve, (b) exact agreement with that mechanism’s prediction (odd n reduces exactly to the $n = 3$ formula) across 13 values of n , and (c) an explicitly identified gap: the multi-lap wraparound argument is not carried out in full rigor the way Log Entry 18’s is. This is reported at that resolution deliberately, per this project’s standing policy (Log Entry 3), rather than rounded up to “proved.”

Log Entry 22: Line-by-Line Rigor Check on the One Complete Proof in This Log

Date: September 22, 2026

What was checked. Before treating Log Entry 18’s theorem as finished, its full proof text (in `verification/prove_syllable_turaev_formula.py`, of which the log’s version is a condensed summary) was given a line-by-line rigor check, hunting specifically for the kind of gap that “feels obvious and isn’t” – unjustified inference steps, mishandled boundary cases, and claims of the form “this composes” or “by symmetry” asserted rather than shown.

Two gaps found, both expository, neither fatal.

1. Lemma 1 is proved for a single syllable given an arbitrary input state; the proof asserted, without spelling out the induction, that this composes across all k syllables of the word. Checking it directly: it does compose, because Lemma 1’s conclusion holds for *arbitrary* input state, so syllable $j + 1$ always sees exactly the frontier state syllable j would have left behind regardless of syllable j ’s length – but this step needed to be said, not left implicit.
2. Lemma 2’s inductive step refers to “the node Z_{k-2} , already in component B ,” which implicitly assumes $k \geq 3$ (Z_{k-2} doesn’t exist for $k = 2$). Checking the

$k = 1 \rightarrow k = 2$ transition directly by hand: it does work, because at $k = 2$ the role of “the node already in B ” is played by TOP_2 itself (the base case of B), not a Z -node – a degenerate instance of the same mechanism, not a different case, but the written proof didn’t say so.

Both gaps were fixed with one clarifying sentence each, directly in the proof script (`verification/prove_syllable_turaev_formula.py`); the argument did not need to be restructured, and the already-passing computational cross-checks (base case $k = 1..40$, 160 random distributions, 203 further (e, k) pairs) required no changes and still pass.

Significance. This closes the loop on Theorem 1 (Log Entry 18): it is not just computationally verified but has now been checked as an argument, line by line, and the gaps that check found were exactly the boundary cases ($k = 1 \rightarrow 2$) this project’s own track record would predict as the likely failure point – consistent with, and reinforcing, the discipline named as this project’s real product in Log Entries 19–21.